

BANKING TRANSACTIONS

End-to-End Power BI Data Analyst Portfolio Report

Portfolio positioning: Data Analyst | SQL | Python | Power BI | Business Intelligence

2.0M Core banking transactions	3.0M Card transactions	95K Accounts	60K Customers
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Purpose

A portfolio-ready case study demonstrating how a Data Analyst converts a multi-domain banking dataset into a decision-oriented Power BI reporting solution. The report covers data preparation, SQL validation, analytical questions, dashboard design, quantified findings, and professional portfolio packaging.

Source artifacts reviewed: banking_transactions.pbix, Banking_Transactions.ipynb, banking_transactionsSQL.sql.

1. Executive Summary

The project combines Python EDA, MySQL analysis and a multi-page Power BI report.

60,000 Customers	95,000 Accounts	2,000,000 Bank transactions	3,000,000 Card transactions
■12.10B Core transaction value	■6,048.38 Avg. core transaction	65,000 Cards	22,000 Loans

Portfolio-level findings

1. The banking dataset is large enough to demonstrate practical BI skills: 2 million core transactions and 3 million card transactions, alongside customer, account, branch, loan, payment, employee and support domains.
2. Customer concentration is visible geographically. Maharashtra accounts for 11,981 customers (19.97%), followed by Madhya Pradesh with 8,059 (13.43%).
3. Account activity is dominated by active accounts: 80,707 of 95,000 (84.96%), while 9,526 are dormant and 4,767 are closed.
4. Deposit and withdrawal transactions are the largest contributors to core transaction value in the EDA, each at approximately ■3.02B.
5. Mobile App is the most frequently used core transaction channel in the notebook analysis, with 333,776 transactions.
6. Card portfolio monitoring is important: 41,185 cards (63.36%) are classified as expired versus 23,815 active in the notebook's expiry-date calculation.
7. The Power BI file is structured as five analytical pages plus a hidden tooltip page: Executive Overview, Customer Analysis, Transaction Analysis, Loans & Credit Risk Analysis, and Credit + Support Analysis.

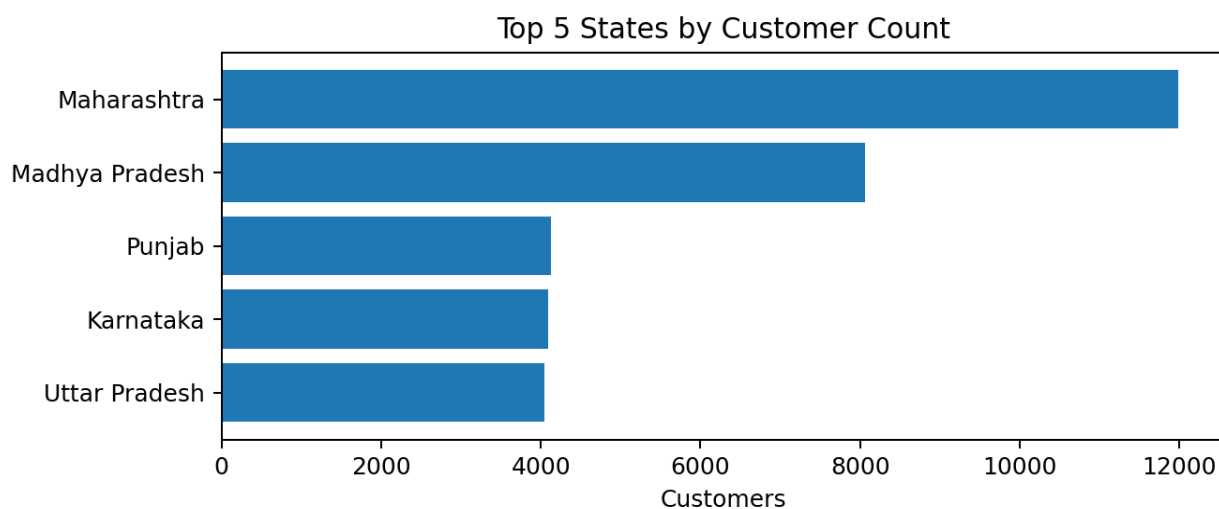
2. Data Landscape & Analytical Architecture

Domain	Rows	Primary analytical purpose
Customers	60,000	Customer profile, geography, income, credit score
Accounts	95,000	Account status, type, balances, branch exposure
Branches	150	Branch footprint and operational comparison
Transactions	2,000,000	Transaction volume, value, type, channel, merchant
Cards	65,000	Card portfolio and status monitoring
Card Transactions	3,000,000	Card spend, merchant categories, fraud flag
Loans	22,000	Loan portfolio, type, amount, status
Loan Payments	600,000	Repayment, principal/interest, late-payment analysis
Employees	1,800	Branch staffing analysis
Support	25,000	Ticket volume, issue type, status, satisfaction

Relational layer: the SQL script defines primary keys and foreign-key relationships across customers/accounts/branches, accounts/transactions, cards/card transactions, loans/loan payments, employees/branches and support/customers. The script also contains an extensive analytical question bank.

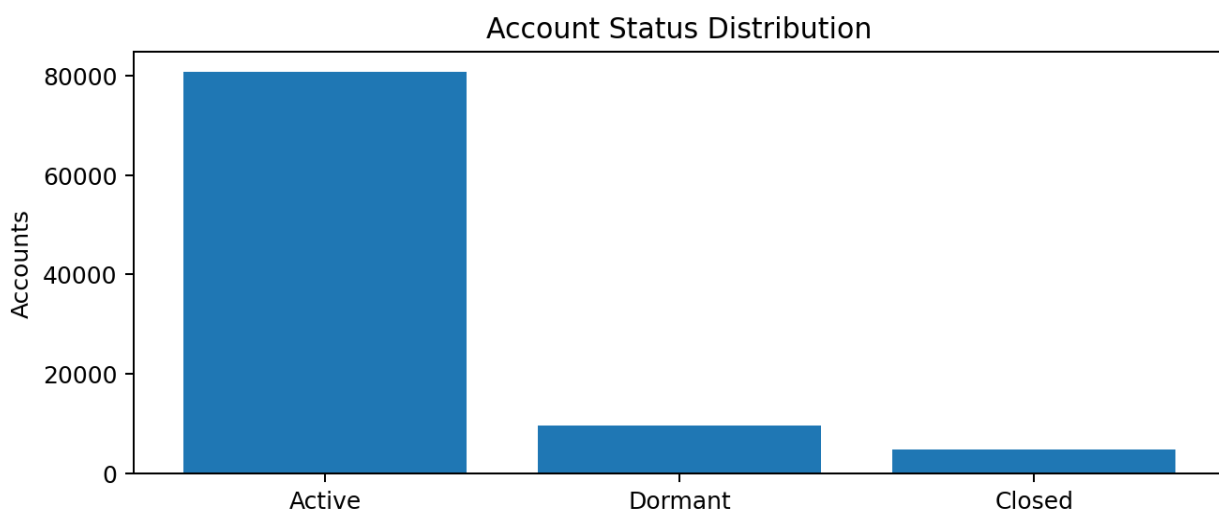
Examples include customer ranking, branch balance, transaction value, high-value transactions, top customers, card spending, loan outstanding estimation and support-ticket analysis.

3. Customer & Account Analysis



Customer distribution is concentrated in Maharashtra and Madhya Pradesh, while the remaining states are comparatively distributed across the network.

19.97%	13.43%	52.24%	45.69%
Maharashtra customer share	Madhya Pradesh share	Male share	Female share



Account activity: 80,707 active, 9,526 dormant and 4,767 closed accounts. Savings accounts are the largest account type with 47,557 accounts (50.06%). Average account balances by type are tightly grouped around ■45K–■47K in the notebook EDA.

4. Transaction Analytics

2.0M

Core transactions

■12.10B

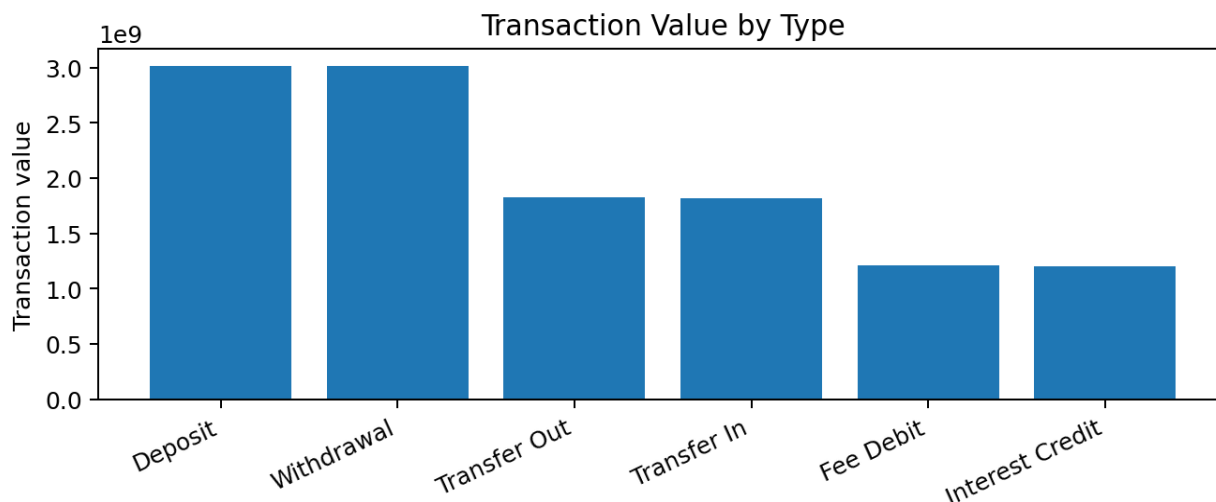
Core transaction value

■6,048.38

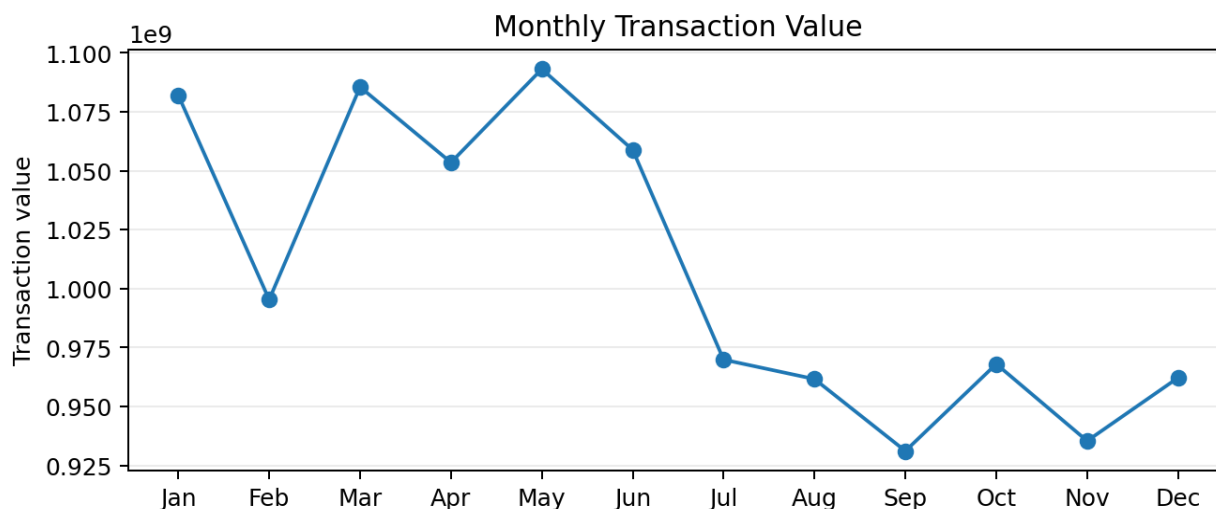
Average transaction

333,776

Mobile App transactions



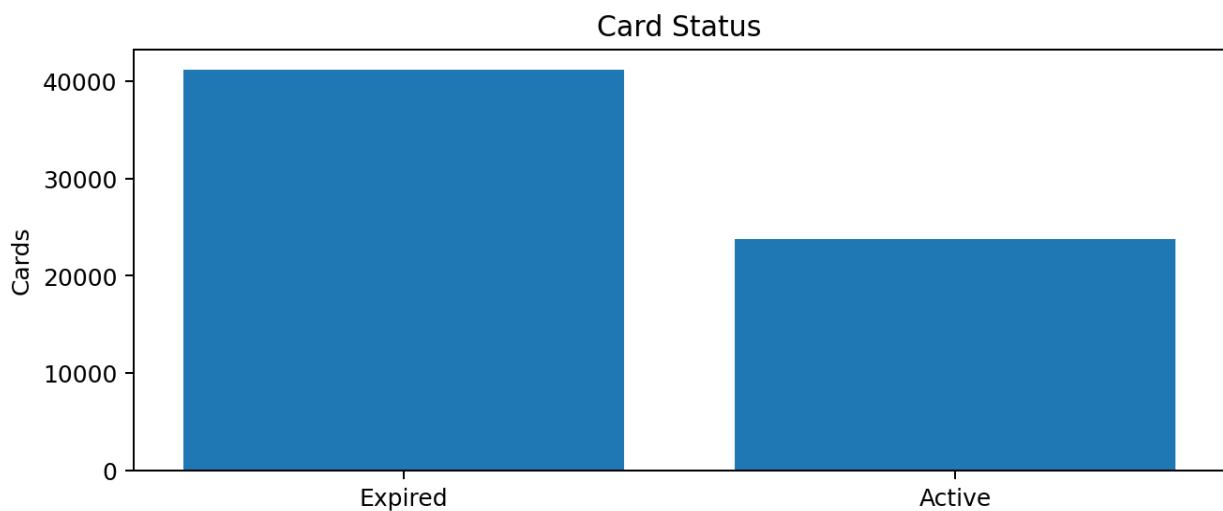
Deposit and withdrawal activity each contribute about ■3.02B, making them the two largest transaction-value categories in the EDA. Transfer In/Out contribute roughly ■1.82B each, while Fee Debit and Interest Credit are around ■1.21B each.



The monthly EDA shows May as the highest transaction-value month at ■1.093B, followed by March at ■1.086B and January at ■1.082B. September is the lowest among the twelve month aggregates at ■0.931B.

The transaction dataset spans January 2018 through June 2026 in the notebook analysis. The 2026 period is partial, so year-over-year comparisons involving 2026 should be treated as incomplete-period comparisons.

5. Cards, Loans & Credit Risk



The card dataset contains 65,000 cards. Based on the notebook's expiry-date logic, 41,185 (63.36%) are expired and 23,815 (36.64%) are active. Debit cards are the largest card type at 35,700.

The loan and payment domains contain 22,000 loans and 600,000 payments. The payment dataset has a 12.0063% late-payment flag rate. Average payment amount is ■10,397.46, while average principal and interest components are ■8,487.30 and ■1,910.16 respectively.

Risk-analysis design: the SQL layer estimates outstanding loan principal as original loan amount less cumulative principal paid, and separately identifies late payments and customers/loans with overdue payments.

Important QA note: the source SQL contains several exploratory queries that should be reviewed before being presented as production KPIs. For example, some queries use LIMIT 1 without an explicit ORDER BY, and a few aliases/conditions contain apparent typing or logic issues. The portfolio report therefore treats the notebook-verified figures as the quantified baseline and the PBIX page structure as the reporting layer.

6. Power BI Report Design

Page	Purpose	Decision questions
Executive Overview	Bank-wide KPI snapshot	How large is the portfolio? What should management focus on?
Customer Analysis	Customer demographics and geography	Who are the customers and where are they concentrated?
Transaction Analysis	Volume, value, channel and merchant behaviour	What drives transaction activity and value?
Loans & Credit Risk Analysis	Loan portfolio and repayment risk	How much is lent, repaid and potentially at risk?
Credit + Support Analysis	Credit profile and service operations	How do credit characteristics and support demand look?
Tooltip	Contextual detail	How can users inspect a selected data point?

Professional dashboard pattern: use a consistent KPI strip, one primary trend visual, one composition visual, one ranking visual, and a concise insight panel. Keep slicers focused on high-value dimensions such as date, branch, account type, transaction type, channel, loan type and customer segment.

Interactivity: the PBIX definition includes enhanced tooltips and a hidden tooltip page, supporting contextual exploration. Drill-through and bookmark patterns can be added where a stakeholder needs guided navigation.

7. End-to-End Data Analyst Workflow

01 | Data ingestion

Loaded banking-domain datasets and inspected structure, types, duplicates and missing values in Python.

02 | Data quality

Validated identifiers, nulls and duplicate IDs; standardized date and numeric fields.

03 | Feature engineering

Created age groups, credit-status categories, transaction month/year fields and card-status logic.

04 | SQL analytics

Built relational keys and analytical queries for customers, accounts, transactions, cards, loans, payments and support.

05 | Power BI modeling

Connected analytical entities into a report-oriented semantic model and organized decision-focused pages.

06 | Dashboarding

Used KPI cards, trends, distributions, rankings, filters and tooltip interactions.

07 | Insight generation

Translated quantitative patterns into operational observations rather than simply presenting charts.

08 | Portfolio packaging

Documented the project for GitHub, LinkedIn and a one-page resume.

8. Business Insights & Analyst Recommendations

Area	Observed pattern	Analyst action
Customer concentration	Maharashtra represents 19.97% of customers and Madhya Pradesh 13.43%	Produce 13-43% segmentation for acquisition, service capacity and ca
Account activity	84.96% of accounts are active, with 10.03% dormant and 5.01% closed	Implement dormant-account monitoring and segment reactivation opportuni
Transaction channels	Mobile App is the most-used channel with 333,776 transactions	Track digital adoption and compare transaction value—not only volum
Transaction mix	Deposits and withdrawals are the largest value categories, but transacti	Use transaction-type trends to understand cash-flow behavior and ide
Card lifecycle	63.36% of cards are classified as expired in the notebook's expiry date	Empty-date expiry pipeline and monitor active/expired/blocked cards by
Credit & support	The project includes credit score, loan repayment, late-payment and se	Centralize support service indicators to identify customers who may ne

Interpretation principle: these are descriptive insights from the supplied project artifacts. They are not causal claims; further validation is required before turning them into operational policy.

9. GitHub / LinkedIn / Resume Packaging

Recommended GitHub structure

banking-transactions-powerbi/

- README.md
- powerbi/banking_transactions.pbix
- sql/banking_transactionsSQL.sql
- python/Banking_Transactions.ipynb
- screenshots/01-executive-overview.png
- screenshots/02-customer-analysis.png
- screenshots/03-transaction-analysis.png
- screenshots/04-loans-credit-risk.png
- screenshots/05-credit-support.png

Resume-ready project entry

Banking Transactions Analytics | Python • MySQL • Power BI

- Analyzed a multi-domain banking dataset covering 60K customers, 95K accounts, 2M core transactions, 65K cards, 3M card transactions and 22K loans.
- Built SQL analytics and Power BI dashboards for customer segmentation, account activity, transaction behavior, credit/loan risk and support operations.
- Identified customer concentration, transaction-channel behavior, account-status mix and card lifecycle patterns using KPI, trend, ranking and drill-down reporting.

LinkedIn project description

Built an end-to-end Banking Transactions Analytics project using Python, MySQL and Power BI. The solution transforms multi-domain banking data into interactive dashboards covering customer insights, account activity, transaction behavior, cards, loans/credit risk and support operations. The project demonstrates data cleaning, exploratory analysis, SQL querying, relational modeling, KPI design, dashboard storytelling and business insight generation.

Recommended screenshots

Capture the five visible PBIX pages at a consistent 16:9 aspect ratio. For GitHub, name them in page order and add one-line captions describing the business question each page answers. For LinkedIn, use the Executive Overview as the first image and place the remaining pages in a carousel.

10. Data Quality & Professional QA Notes

Check	Result / note
Dataset scale	Verified from the notebook: 60K customers, 95K accounts, 150 branches, 2M transactions, 65K cards, 3M card transaction
Identifiers	Notebook checks show no duplicate card transaction IDs, loan payment IDs or support ticket IDs in the inspected datasets;
Date coverage	Core transactions run from 2018-01-01 through 2026-06-30 in the notebook output. 2026 is partial.
Credit status	The notebook creates credit-status bands using score bins; the SQL script also contains a separate CASE-based classifica
SQL QA	Review queries with missing ORDER BY clauses, questionable aliases, or join conditions before using their outputs as pro
Power BI QA	Validate all KPI cards against the SQL/Python baseline after refresh, especially totals, ratios, date filters and relationship-d

Final portfolio assessment

The project demonstrates an end-to-end analyst workflow: raw banking data → Python EDA → relational SQL analysis → Power BI semantic/report layer → decision-oriented dashboards → portfolio documentation. The next improvement is to tighten metric definitions, validate every Power BI KPI against a single source of truth, and add concise insight annotations to each dashboard page.

Source artifacts: supplied Power BI PBIX, Python notebook and MySQL SQL script. Quantified figures in this report are limited to values verifiable from those supplied artifacts.